

Chapter 1: Discrete-Time Signals & Systems - Fundamentals

1.1 Introduction to Discrete-Time Signals and Systems

A discrete-time signal $x[n]$ is defined for integer values of n in $(-\infty, \infty)$. A discrete-time system T transforms an input sequence $x[n]$ into an output sequence $y[n] = T\{x[n]\}$.

Key System Properties:

1. Linearity: The system satisfies superposition: $T\{a \cdot x_1[n] + b \cdot x_2[n]\} = a \cdot T\{x_1[n]\} + b \cdot T\{x_2[n]\}$.
2. Time-Invariance: A time-shift in the input causes an identical time-shift in the output: $T\{x[n - k]\} = y[n - k]$.
3. Causality: The output $y[n]$ at any index n depends only on present and past inputs $x[k]$ for $k \leq n$, and not on future inputs.
4. Stability (BIBO): A discrete-time LTI system is bounded-input bounded-output (BIBO) stable if and only if its impulse response $h[n]$ is absolutely summable:
$$\sum_{k=-\infty}^{\infty} |h[k]| < \infty.$$

If the impulse response is absolutely summable, any bounded input $|x[n]| \leq B_x$ produces a bounded output $|y[n]| \leq B_y < \infty$.

Chapter 1: Difference Equations & Convolution

1.2 Linear Constant-Coefficient Difference Equations (LCCDE)

An N-th order LCCDE describes an LTI system relating input $x[n]$ and output $y[n]$:

$$\sum_{k=0}^N a_k y[n - k] = \sum_{m=0}^M b_m x[n - m], \text{ with } a_0 \neq 0.$$

Linear Convolution Representation:

For any LTI system, the response to an arbitrary input $x[n]$ is given by the convolution sum:

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k].$$

Commutative Property: $x[n] * h[n] = h[n] * x[n]$.

Associative Property: $(x[n] * h_1[n]) * h_2[n] = x[n] * (h_1[n] * h_2[n])$.

Distributive Property: $x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$.

Chapter 2: Discrete-Time Fourier Transform (DTFT)

2.1 DTFT Definition and Existence

The Discrete-Time Fourier Transform (DTFT) of a sequence $x[n]$ is defined as:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] * e^{-j\omega n}$$

Inverse DTFT (IDTFT):

$$x[n] = (1 / 2\pi) * \int_{-\pi}^{\pi} X(e^{j\omega}) * e^{j\omega n} d_{\omega}$$

Existence Condition: The DTFT converges uniformly if $x[n]$ is absolutely summable:

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty.$$

Periodicity: $X(e^{j\omega})$ is strictly periodic with fundamental period 2π :

$$X(e^{j(\omega + 2\pi)}) = X(e^{j\omega}).$$

2.2 Mathematical Properties of the DTFT

1. Linearity: $a \cdot x_1[n] + b \cdot x_2[n] \leftrightarrow a \cdot X_1(e^{j\omega}) + b \cdot X_2(e^{j\omega})$.
2. Time Shifting: $x[n - n_0] \leftrightarrow e^{-j\omega n_0} \cdot X(e^{j\omega})$.
3. Frequency Shifting: $e^{j\omega_0 n} \cdot x[n] \leftrightarrow X(e^{j(\omega - \omega_0)})$.
4. Conjugation & Symmetry: For a real-valued sequence $x[n]$:
 $X(e^{-j\omega}) = X^*(e^{j\omega})$
 $|X(e^{-j\omega})| = |X(e^{j\omega})|$ (Magnitude is an even function)
 $\arg\{X(e^{-j\omega})\} = -\arg\{X(e^{j\omega})\}$ (Phase is an odd function)
5. Parseval's Theorem:
 $\sum_{n=-\infty}^{\infty} |x[n]|^2 = (1 / 2\pi) \cdot \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$.

Chapter 3: The Z-Transform & Region of Convergence

3.1 Definition and Region of Convergence (ROC)

The two-sided Z-transform of a discrete-time sequence $x[n]$ is defined as:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] \cdot z^{-n}, \text{ where } z = r \cdot e^{j\omega}.$$

Properties of the ROC:

1. The ROC of $X(z)$ consists of a ring in the z -plane centered at the origin: $r_R < |z| < r_L$.
2. The ROC cannot contain any poles of $X(z)$.
3. For a finite-duration sequence, the ROC is the entire z -plane, except possibly $z=0$ or $z=\infty$.
4. For a right-sided sequence, the ROC extends outward from the outermost pole to infinity ($|z| > r_{\max}$).
5. For a left-sided sequence, the ROC extends inward from the innermost pole to the origin ($|z| < r_{\min}$).
6. For a two-sided sequence, the ROC is an open annular ring between two concentric circles.

3.2 Inversion Techniques

1. Inspection Method: Recognizing standard transform pairs from lookup tables.

2. Partial Fraction Expansion (PFE):

If $X(z) = B(z)/A(z)$ is a rational function with distinct poles p_k :

$X(z)/z = \sum_{k=1}^N (A_k / (z - p_k))$, where $A_k = [(z - p_k) * (X(z)/z)]_{z=p_k}$.

3. Power Series Expansion (Long Division):

Expanding $X(z)$ in ascending or descending powers of z according to the causality specified by the ROC.

4. Contour Integration: Using Cauchy Residue Theorem:

$x[n] = (1 / 2\pi j) * \text{contour_integral } X(z) * z^{n-1} dz = \text{sum residues.}$

Chapter 4: LTI Systems in the Z-Domain

4.1 Transfer Function $H(z)$, Poles, and Zeros

The system transfer function $H(z)$ is the Z-transform of the impulse response $h[n]$:

$$H(z) = Y(z) / X(z) = \sum_{m=0}^M b_m z^{-m} / \sum_{k=0}^N a_k z^{-k}.$$

Stability Condition:

An LTI system is BIBO stable if and only if the Region of Convergence of $H(z)$ includes the unit circle ($|z| = 1$).

Causal and Stable Systems:

For a causal LTI system, all poles of $H(z)$ must strictly lie inside the unit circle ($|p_k| < 1$ for all k). If any pole lies on or outside the unit circle, the causal system is unstable or marginally stable.

Chapter 4: Special Filter Classes

4.2 Minimum-Phase and All-Pass Systems

Minimum-Phase System:

A causal, stable LTI system $H(z)$ is called minimum-phase if all its poles AND all its zeros lie strictly inside the unit circle ($|z| < 1$). The inverse filter $1/H(z)$ is also causal and stable.

All-Pass System:

An all-pass filter has a constant magnitude response across all frequencies: $|H_{ap}(e^{j\omega})| = 1$ for all ω .

Its poles and zeros occur in conjugate reciprocal pairs: a pole at $z = p$ is matched with a zero at $z = 1/p^*$.

Any causal, stable transfer function can be uniquely decomposed as $H(z) = H_{min}(z) * H_{ap}(z)$.

Chapter 5: Discrete Fourier Transform (DFT)

5.1 Discrete Fourier Transform Definition

The N-point DFT of a finite-length sequence $x[n]$, $0 \leq n \leq N-1$, is given by:

$$X[k] = \sum_{n=0}^{N-1} x[n] * W_N^{k*n}, \text{ for } k = 0, 1, \dots, N-1,$$
where $W_N = e^{-j*(2*\pi/N)}$ is the twiddle factor.

Inverse DFT (IDFT):

$$x[n] = (1 / N) * \sum_{k=0}^{N-1} X[k] * W_N^{-k*n}, \text{ for } n = 0, 1, \dots, N-1.$$

Relation to DTFT:

The DFT corresponds to uniformly spaced frequency samples of the DTFT evaluated at $\omega_k = (2*\pi*k)/N$.

Chapter 5: Circular vs Linear Convolution

5.2 Circular Convolution and Frequency Domain Filtering

Circular Convolution Definition:

The circular convolution of two N-point sequences $x_1[n]$ and $x_2[n]$ is defined as:

$$x_3[n] = \sum_{m=0}^{N-1} x_1[m] * x_2[(n - m)_N], \text{ where } ((n))_N \text{ denotes modulo-}N \text{ indexing.}$$

Property: $\text{DFT}\{x_1[n] \circledast x_2[n]\} = X_1[k] * X_2[k]$.

Linear Convolution via Circular Convolution:

If $x_1[n]$ has length L and $x_2[n]$ has length M, their linear convolution has length $L + M - 1$.

To compute linear convolution using the DFT without time-domain aliasing, both sequences must be zero-padded to a length $N \geq L + M - 1$ before computing the N-point DFT and IDFT.

Chapter 6: Fast Fourier Transform (FFT)

6.1 Radix-2 Decimation-in-Time (DIT) FFT

The Cooley-Tukey Radix-2 DIT FFT recursively divides an N-point DFT into two N/2-point DFTs of even-indexed and odd-indexed samples:

$$X[k] = G[k] + W_N^k * H[k]$$

$$X[k + N/2] = G[k] - W_N^k * H[k], \text{ for } k = 0, 1, \dots, N/2 - 1.$$

Computational Complexity Comparison:

- Direct N-point DFT: N^2 complex multiplications, $N(N-1)$ complex additions.
- Radix-2 DIT FFT: $(N/2) * \log_2(N)$ complex multiplications, $N * \log_2(N)$ complex additions.

For $N = 1024$, direct DFT requires 1,048,576 complex multiplications, whereas FFT requires only 5,120 multiplications.

Chapter 6: Decimation-in-Frequency (DIF) FFT

6.2 Radix-2 Decimation-in-Frequency (DIF) Algorithm

In DIF FFT, the time-domain sequence is split into the first half and second half of samples:

$$x_first[n] = x[n] + x[n + N/2]$$

$$x_second[n] = (x[n] - x[n + N/2]) * W_N^n, \text{ for } n = 0, 1, \dots, N/2 - 1.$$

Bit Reversal Permutation:

For DIT FFT, the inputs must be bit-reversed while outputs appear in normal natural order.

For DIF FFT, the inputs are in natural order while outputs emerge in bit-reversed order.

Chapter 7: Fundamentals of Digital Filters

7.1 Filter Specifications and Ideal Frequency Responses

Digital filters process discrete-time signals to pass desired frequency bands and attenuate undesired bands.

Standard Specifications for a Lowpass Filter:

- Passband edge frequency: ω_p
- Stopband edge frequency: ω_s
- Passband ripple: $1 - \delta_p \leq |H(e^{j\omega})| \leq 1 + \delta_p$ (or α_p in dB)
- Stopband attenuation: $|H(e^{j\omega})| \leq \delta_s$ (or α_s in dB)
- Transition band: $\Delta\omega = \omega_s - \omega_p$.

Chapter 8: Finite Impulse Response (FIR) Filters

8.1 Linear Phase FIR Filter Characteristics

FIR filters possess an impulse response $h[n]$ of finite duration M (length $N = M + 1$):

$$H(z) = \sum_{n=0}^M h[n] z^{-n}.$$

Linear Phase Condition:

An FIR filter exhibits generalized linear phase if its impulse response satisfies symmetry or antisymmetry:

$$h[n] = h[M - n] \text{ (Symmetric, Linear phase with constant group delay } \tau = M/2)$$

$$h[n] = -h[M - n] \text{ (Antisymmetric, 90-degree constant phase shift).}$$

Four Types of Linear Phase FIR Filters:

- Type I: Symmetric, M is even (N odd). Can implement lowpass, highpass, bandpass, bandstop.
- Type II: Symmetric, M is odd (N even). Cannot implement highpass or bandstop because $H(e^{j\pi}) = 0$.
- Type III: Antisymmetric, M is even. $H(e^{j0}) = H(e^{j\pi}) = 0$. Suitable for bandpass/differentiator.
- Type IV: Antisymmetric, M is odd. $H(e^{j0}) = 0$. Suitable for highpass and differentiator.

Chapter 8: FIR Window Design Method

8.2 Window Functions in FIR Design

The window method truncates the ideal infinite impulse response $h_d[n]$ by multiplying with a finite window $w[n]$:

$$h[n] = h_d[n] * w[n], \text{ for } 0 \leq n \leq M.$$

Window Comparison Table:

1. Rectangular: Mainlobe width = $4\pi/M$, Peak sidelobe = -13 dB, Min stopband attenuation = 21 dB.
2. Bartlett (Triangular): Mainlobe width = $8\pi/M$, Peak sidelobe = -25 dB, Min stopband = 25 dB.
3. Hanning: Mainlobe width = $8\pi/M$, Peak sidelobe = -31 dB, Min stopband = 44 dB.
4. Hamming: Mainlobe width = $8\pi/M$, Peak sidelobe = -43 dB, Min stopband = 53 dB.
5. Blackman: Mainlobe width = $12\pi/M$, Peak sidelobe = -58 dB, Min stopband = 74 dB.

Trade-off: As sidelobe attenuation increases, mainlobe width widens, yielding a wider transition bandwidth.

8.3 Kaiser Parameterized Window Formulation

The Kaiser window allows independent adjustment of transition bandwidth and stopband attenuation:

$w[n] = I_0(\beta \sqrt{1 - (2n/M - 1)^2}) / I_0(\beta)$, $0 \leq n \leq M$,
where $I_0(x)$ is the modified zero-th order Bessel function of the first kind.

Empirical Formulas for Kaiser Parameters:

Given stopband attenuation $A = -20 \cdot \log_{10}(\delta)$:

- For $A > 50$ dB: $\beta = 0.1102 \cdot (A - 8.7)$.
- For $21 \leq A \leq 50$ dB: $\beta = 0.5842 \cdot (A - 21)^{0.4} + 0.07886 \cdot (A - 21)$.
- Filter length: $M = (A - 8) / (2.285 \cdot \Delta\omega)$.

Chapter 8: Frequency Sampling FIR Design

8.4 Frequency Sampling Method

In frequency sampling design, the desired continuous frequency response $H_d(e^{j\omega})$ is sampled at N points:

$$H[k] = H_d(e^{j(2\pi k/N)}), \quad k = 0, 1, \dots, N-1.$$

The filter coefficients are obtained via inverse DFT:

$$h[n] = (1/N) * \sum_{k=0}^{N-1} H[k] * e^{j(2\pi kn/N)}.$$

Optimization: Introducing unconstrained transition band samples significantly improves stopband attenuation.

Chapter 9: Infinite Impulse Response (IIR) Filters

9.1 Introduction to IIR Filter Design

IIR filters have infinite-duration impulse response and can achieve very sharp cutoff characteristics with significantly lower filter order compared to FIR filters.

Transfer function:

$$H(z) = \frac{\sum_{m=0}^M b_m z^{-m}}{(1 + \sum_{k=1}^N a_k z^{-k})}$$

Standard Analog Filter Prototypes:

1. Butterworth: Maximally flat magnitude response in passband and stopband.
2. Chebyshev Type I: Equiripple passband, monotonic stopband.
3. Chebyshev Type II (Inverse Chebyshev): Monotonic passband, equiripple stopband.
4. Elliptic (Cauer): Equiripple in both passband and stopband, lowest order for given specifications.

Chapter 9: Impulse Invariance Method

9.2 Impulse Invariance IIR Filter Design

The impulse invariance method designs digital filter $h[n]$ by sampling the continuous impulse response $h_a(t)$:
 $h[n] = T * h_a(nT)$, where T is the sampling interval.

Mapping Relationship:

A pole at $s = p_k$ in the s -plane maps directly to a pole at $z = e^{p_k * T}$ in the z -plane.

Limitation - Aliasing:

The digital frequency response is an aliased summation of the analog frequency response:

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} H_a(j(\omega/T + 2\pi k/T)).$$

Impulse invariance cannot be used for highpass or bandstop filters due to severe aliasing.

Chapter 9: Bilinear Transformation Method

9.3 Bilinear Transformation (BLT)

The bilinear transformation converts an analog transfer function $H_a(s)$ to a digital transfer function $H(z)$ using the trapezoidal numerical integration substitution:

$$s = (2/T) * ((1 - z^{-1}) / (1 + z^{-1})) = (2/T) * ((z - 1) / (z + 1)).$$

Key Properties:

1. The entire left-half of the s-plane ($\text{Re}\{s\} < 0$) maps strictly inside the unit circle ($|z| < 1$), guaranteeing that a stable analog filter transforms into a stable digital filter.
2. The $j\Omega$ imaginary axis in the s-plane maps one-to-one onto the unit circle $z = e^{j\omega}$.
3. Completely eliminates aliasing because the infinite analog frequency axis is compressed into $[-\pi, \pi]$.

Chapter 9: Frequency Warping & Prewarping

9.4 Frequency Warping in Bilinear Transformation

Substituting $s = j\Omega$ and $z = e^{j\omega}$ into the bilinear transform yields:

$$j\Omega = \frac{2}{T} \cdot \frac{(e^{j\omega} - 1)}{(e^{j\omega} + 1)} = j \cdot \frac{2}{T} \cdot \tan(\omega / 2).$$

Therefore, the non-linear relationship between analog frequency Ω and digital frequency ω is:

$$\Omega = \frac{2}{T} \cdot \tan(\omega / 2), \text{ and inversely: } \omega = 2 \cdot \arctan(\Omega \cdot T / 2).$$

Frequency Prewarping Rule:

Before applying analog design tables, the digital critical frequencies (ω_p , ω_s) must be prewarped to analog frequencies:

$$\Omega_p = \frac{2}{T} \cdot \tan(\omega_p / 2)$$

$$\Omega_s = \frac{2}{T} \cdot \tan(\omega_s / 2).$$

Chapter 10: Butterworth Lowpass Filter Design

10.1 Butterworth Filter Equations and Order Determination

The magnitude-squared response of an N-th order analog Butterworth filter is:

$$|H(j\Omega)|^2 = 1 / (1 + (\Omega / \Omega_c)^{2N}).$$

Design Formula for Filter Order N:

Given passband attenuation α_p (dB) at Ω_p and stopband attenuation α_s (dB) at Ω_s :

$$N = \text{ceil} \left(\frac{\log_{10}((10^{0.1\alpha_s} - 1) / (10^{0.1\alpha_p} - 1))}{2 * \log_{10}(\Omega_s / \Omega_p)} \right).$$

Cutoff Frequency Ω_c :

$$\Omega_c = \Omega_p / ((10^{0.1\alpha_p} - 1)^{1 / (2N)}).$$

Poles of the Butterworth filter lie symmetrically on a circle of radius Ω_c in the left-half s-plane:

$$s_k = \Omega_c * e^{j\pi * (2k + N - 1) / (2N)}, \text{ for } k = 1, 2, \dots, N.$$

Chapter 10: Chebyshev Lowpass Filter Design

10.2 Chebyshev Type I Filter Specifications

The magnitude response of a Chebyshev Type I filter is:

$$|H(j\Omega)|^2 = 1 / (1 + \epsilon^2 * C_N^2(\Omega / \Omega_p)),$$

where $C_N(x)$ is the N-th order Chebyshev polynomial:

$$C_N(x) = \cos(N * \arccos(x)) \text{ for } |x| \leq 1, \text{ and } \cosh(N * \operatorname{arccosh}(x)) \text{ for } |x| > 1.$$

Ripple parameter epsilon:

$$\epsilon = \sqrt{10^{0.1 * \alpha_p} - 1}.$$

Filter Order N Calculation:

$$N = \lceil \operatorname{arccosh}(\sqrt{(10^{0.1 * \alpha_s} - 1) / \epsilon^2}) / \operatorname{arccosh}(\Omega_s / \Omega_p) \rceil.$$

Chapter 11: Realization Structures for Digital Filters

11.1 Direct Form Realizations for IIR Filters

Consider transfer function $H(z) = (b_0 + b_1 z^{-1} + b_2 z^{-2}) / (1 + a_1 z^{-1} + a_2 z^{-2})$.

1. Direct Form I (DF-I):

Implements the difference equation directly: computes zeros followed by poles. Requires $M + N$ delay elements.

2. Direct Form II (DF-II) Canonical Form:

Computes intermediate variable $w[n] = x[n] - a_1 w[n-1] - a_2 w[n-2]$, followed by $y[n] = b_0 w[n] + b_1 w[n-1] + b_2 w[n-2]$.

Requires only $\max(M, N)$ delay elements (canonic in delay elements).

Chapter 11: Modular Realizations

11.2 Cascade and Parallel Form Realizations

Cascade Realization:

High-order $H(z)$ is factored into a product of second-order biquad sections:

$$H(z) = b_0 * \prod_{k=1}^K ((1 + b_{1k} z^{-1} + b_{2k} z^{-2}) / (1 + a_{1k} z^{-1} + a_{2k} z^{-2})).$$

Minimizes sensitivity of pole locations to coefficient quantization errors.

Parallel Realization:

Obtained using partial fraction expansion as a sum of second-order sections plus a constant:

$$H(z) = C + \sum_{k=1}^K ((\gamma_{0k} + \gamma_{1k} z^{-1}) / (1 + a_{1k} z^{-1} + a_{2k} z^{-2})).$$

12.1 Quantization Noise and Limit Cycle Oscillations

In fixed-point hardware implementations, finite precision causes three primary artifacts:

1. Coefficient Quantization: Shifts pole and zero locations; in high-order direct form filters, poles may move outside the unit circle causing instability.
2. Roundoff Noise: Product roundoff introduces white noise variance $\sigma_e^2 = 2^{-2B} / 12$ at each multiplier.
3. Limit Cycle Oscillations:
 - Zero-Input Limit Cycles: Due to non-linear quantization rounding in recursive loops, producing sustained periodic output even when input is zero.
 - Overflow Oscillations: Due to two's complement wrap-around overflow. Eliminated using saturation arithmetic logic.